

Technical Project Report

Paper Summary:

"The Martynal Value of Adaptive Gradient Methods in Machine Learning" by Ashia C. Wilson, Rebecca Roelofs, Mitchell Stern, Nathan Srebro, and Benjamin Recht analyze the effectiveness of optimizer algorithms like Adam and RMSProp relative to other algorithms such as Stochastic Gradient Descent (SGD). The authors found that while optimizers are great for quick convergence, they ultimately fall short on actual performance on broad data. Using a thorough experiment as evidence, the authors challenge the conventional use of optimizer algorithms.

Experimental Setup

This project reproduces the experimental philosophy presented in the aforementioned research by evaluating the practical differences between adaptive and non-adaptive optimization methods on image classification. The core objective was to compare Adam and SGD (with momentum) on the "Fashion-MNIST, "Zalando Image classification dataset" benchmark using both a manual NumPy implementation and PyTorch implementation.

The dataset consists of 70,000 grayscale images (28×28 pixels) across 10 clothing categories. The training split contained 60,000 images, which were further divided into 50,000 training samples and 10,000 validation samples. The test set consisted of 10,000 images. All pixel values were normalized to the range $[0, 1]$ by dividing by 255.

The model architecture used throughout the main experiment was a two-layer multilayer perceptron (MLP):

- Input layer: 784 neurons
(flattened image)
- Hidden layer: 256 neurons
- Activation: ReLU
- Output layer: 10 neurons
(softmax via cross-entropy)

The NumPy implementation manually implemented forward propagation, backpropagation, gradient computation, and parameter updates.

The PyTorch implementation used the same architecture but relied on autograd for gradient computation. Both implementations performed learning rate grid search.

Learning rate search space:

Adam: $[1e^{-4}, 3e^{-4}, 1e^{-3}, 3e^{-3}]$

SGD: $[1e^{-3}, 1e^{-2}, 5e^{-2}, 1e^{-1}]$

Each experiment was repeated across five random seeds to measure stability and compute 95% confidence intervals.

Results

The primary finding was that both Adam and SGD achieved very similar final test accuracies, despite differing optimization behavior during training.

Main Observations:

- Adam converged significantly faster in early epochs.
- SGD required more iterations but often matched or slightly exceeded Adam in final performance.
- Confidence intervals were narrow, indicating stable optimization.

This supports the original paper's central argument: adaptive methods optimize faster, but do not necessarily generalize better.

Discrepancy Analysis

Although the reproduced trends matched the original research paper, exact numerical results differed. Several factors explain this discrepancy:

1. Different dataset - The original paper evaluated larger and more complex datasets (e.g., CIFAR and language modeling benchmarks), while this project used Fashion-MNIST. Fashion-MNIST is simpler and lower-dimensional, reducing optimization difficulty.
2. Different Model Architecture - The original work evaluated deeper architectures and more complex optimization landscapes. This project used a relatively shallow MLP.

3. Hyperparameter Search
Scope - The original paper performed broader tuning across more parameters, while this project focused primarily on learning rate tuning.

Despite these differences, the qualitative behavior remained consistent.

Ablation

The ablation study showed that optimizer hyperparameters only had a small effect on performance compared to architectural changes. The best optimizer variant was Adam $\beta_1 = 0.5$, achieving $89.99\% \pm 0.11\%$, slightly outperforming the baseline. Mild weight decay also improved both Adam and SGD, while extreme SGD momentum values (0.0 and 0.99) significantly reduced accuracy. This suggests that optimizer tuning can improve stability, but gains remain relatively limited.

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In contrast, MLP architecture had a much larger impact on performance. The strongest model was the wide-deep architecture (512-512-256) with $90.49\% \pm 0.19\%$, followed by larger hidden layers and deeper networks. Dropout also improved generalization, while smaller models reduced accuracy. Zero initialization completely failed (0% accuracy), confirming the importance of proper weight initialization. Overall, the ablation results show that architecture and regularization mattered more than optimizer choice in this experiment.

Reflection

This project demonstrated that optimizer choice affects optimization speed more than final generalization performance.

Key takeaways:

- Adam converges faster
- SGD remains highly competitive in final accuracy.
- Model architecture matters more than optimizer selection.

Future work includes testing the NumPy and PyTorch implementations on different datasets.

AI Use Disclosure

AI assistance was used during this project to support development, debugging, and technical writing.

Specifically, AI was used for:

- Code structure refinement
- Debugging training loop issues
- Improving experiment organization
- Generating visualization recommendations
- Assisting with interpretation of experimental results
- Drafting technical summaries and support sections.

All algorithmic implementations, experiment decisions, ablation design, and result validation were reviewed and verified manually.